Stage 5 Report: Final Evaluation and Analysis

Image Super-Resolution with Deep Generative Models · Computer Vision 2026 · 3 Nov 2026 Team: Iuliia Kurnaeva, Alena Petrenko, Amaliya Kharisova, Dmitrii Kuznetsov

Repository: https://github.com/Ephy01/Image-Super-Resolution-with-Deep-Generative-Models

1. Completed experiments

Four pretrained methods ran on every input: bicubic, Real-ESRNet, Real-ESRGAN and ResShift v3 (4 steps). Nothing is trained. The test split has 276 images (80 objects, 100 faces, 96 text), downsampled ×4 to 128×128 and compressed at JPEG none, 50, 30 and 10: 1,104 inputs per method. Dev has 64 images.

- **Hardware.** Tesla T4, batch size 1. Real-ESRNet and Real-ESRGAN use float32; ResShift uses float32 weights with float16 autocast. Bicubic runs on the notebook CPU. Metrics ran on an Apple M4 CPU.
- **Frozen settings.** `configs/final.yaml` (SHA-256 `70f6c341`) was frozen before the test run from dev HR images only. OCR: Tesseract 5.5.1, `--psm 4 --oem 1`, the best of 18 candidates, HR CER 0.017 on dev. ArcFace: buffalo_l with a 512 px detector input (640 px missed 12 of 20 dev faces); match threshold 0.182, the 1% false match rate on 190 dev impostor pairs. MUSIQ: KonIQ-10k checkpoint. ResShift: seed offsets 0 to 4. Intervals: 95% percentile bootstrap [lo, hi], 10,000 resamples, seed 42, no multiple-comparison correction. The analysis and failure-selection rules are in the same file.
- **Reproduction.** With Pillow's BASIC text layout, 320 of 320 dev inputs equal the stage 4 archive. Bicubic, Real-ESRNet and Real-ESRGAN reproduce the archived per-image metrics within tolerance: at most 0.0049 dB PSNR, 0.0002 SSIM and 0.0005 LPIPS. ResShift does not: at most 0.032 dB PSNR (18 of 256 values above 0.01 dB) and 0.0013 LPIPS; the mean absolute PSNR difference is 0.0035 dB. The inputs and weights match; the model computation is unchanged. Stage 4 used CUDA 13.0 and this run CUDA 12.8, which probably explains the difference.

2. Final results (test split, 276 images)

PSNR and SSIM on the Y channel with a 4-pixel border crop; LPIPS with AlexNet; MUSIQ is no-reference. Last 2 columns: CER for text; ArcFace similarity to the HR face and match rate for faces. Means over images; best per content and condition in bold.

No JPEG

Content	Method	PSNR ↑	SSIM ↑	LPIPS ↓	MUSIQ ↑	CER ↓ / ArcFace sim ↑	Match rate ↑
Objects	Bicubic	26.69	0.718	0.465	33.1	n/a	n/a
Objects	Real-ESRNet	**27.54**	**0.765**	0.357	58.1	n/a	n/a
Objects	Real-ESRGAN	25.65	0.709	0.244	**66.4**	n/a	n/a
Objects	ResShift	26.56	0.731	**0.192**	65.7	n/a	n/a
Objects	HR reference	n/a	n/a	n/a	68.6	n/a	n/a
Faces	Bicubic	32.08	0.869	0.296	41.5	**0.983**	**1.00**
Faces	Real-ESRNet	**32.68**	**0.886**	0.245	62.3	0.977	1.00
Faces	Real-ESRGAN	30.25	0.852	0.166	71.5	0.954	1.00
Faces	ResShift	31.32	0.858	**0.113**	**72.4**	0.972	1.00
Faces	HR reference	n/a	n/a	n/a	73.6	n/a	n/a
Text	Bicubic	16.42	0.706	0.436	35.5	**0.210**	n/a
Text	Real-ESRNet	19.25	0.838	0.184	49.4	0.220	n/a
Text	Real-ESRGAN	17.79	0.819	0.107	**55.0**	0.340	n/a
Text	ResShift	**19.72**	**0.859**	**0.086**	54.0	0.243	n/a
Text	HR reference	n/a	n/a	n/a	57.4	0.019	n/a

JPEG 10

Content	Method	PSNR ↑	SSIM ↑	LPIPS ↓	MUSIQ ↑	CER ↓ / ArcFace sim ↑	Match rate ↑
Objects	Bicubic	23.52	0.570	0.758	19.6	n/a	n/a
Objects	Real-ESRNet	**23.85**	**0.609**	0.595	40.7	n/a	n/a
Objects	Real-ESRGAN	22.67	0.571	0.440	**60.5**	n/a	n/a
Objects	ResShift	22.98	0.552	**0.434**	56.7	n/a	n/a
Objects	HR reference	n/a	n/a	n/a	68.6	n/a	n/a
Faces	Bicubic	27.38	0.754	0.642	20.4	**0.689**	0.96
Faces	Real-ESRNet	**27.83**	**0.789**	0.453	36.9	0.656	**1.00**
Faces	Real-ESRGAN	26.70	0.762	0.340	56.7	0.608	1.00
Faces	ResShift	27.19	0.755	**0.318**	**60.5**	0.611	1.00
Faces	HR reference	n/a	n/a	n/a	73.6	n/a	n/a
Text	Bicubic	15.09	0.579	0.642	24.2	0.584	n/a
Text	Real-ESRNet	**15.62**	**0.689**	0.409	40.2	**0.574**	n/a
Text	Real-ESRGAN	14.35	0.680	0.215	**52.9**	0.594	n/a
Text	ResShift	14.95	0.668	**0.195**	50.5	0.624	n/a
Text	HR reference	n/a	n/a	n/a	57.4	0.019	n/a

All JPEG levels with intervals: `summary_test.csv`; CER by font size: `summary_text_by_size_test.csv`.

3. Comparison

Paired differences on the same images, method minus Real-ESRNet, 95% intervals [lo, hi]:

Content	JPEG	Method (minus Real-ESRNet)	PSNR (dB)	LPIPS	MUSIQ	CER / ArcFace sim
Objects	none	Real-ESRGAN	-1.89 [-2.06, -1.73]	-0.113 [-0.129, -0.098]	+8.3 [7.0, 9.7]	n/a
Objects	none	ResShift	-0.98 [-1.20, -0.77]	-0.164 [-0.181, -0.148]	+7.6 [6.4, 8.9]	n/a
Faces	none	Real-ESRGAN	-2.43 [-2.56, -2.30]	-0.080 [-0.085, -0.074]	+9.3 [8.6, 10.0]	-0.023 [-0.025, -0.021]
Faces	none	ResShift	-1.36 [-1.52, -1.21]	-0.133 [-0.139, -0.126]	+10.1 [9.3, 10.9]	-0.004 [-0.006, -0.003]
Text	none	Real-ESRGAN	-1.47 [-1.59, -1.34]	-0.077 [-0.095, -0.061]	+5.6 [5.0, 6.3]	+0.120 [0.094, 0.146]
Text	none	ResShift	+0.47 [0.36, 0.58]	-0.098 [-0.119, -0.078]	+4.6 [3.8, 5.5]	+0.023 [0.013, 0.033]
Objects	10	Real-ESRGAN	-1.18 [-1.28, -1.08]	-0.155 [-0.172, -0.138]	+19.8 [18.1, 21.5]	n/a
Objects	10	ResShift	-0.86 [-1.02, -0.72]	-0.161 [-0.181, -0.141]	+16.0 [13.9, 18.1]	n/a
Faces	10	Real-ESRGAN	-1.13 [-1.20, -1.06]	-0.112 [-0.118, -0.106]	+19.8 [18.3, 21.3]	-0.048 [-0.054, -0.041]
Faces	10	ResShift	-0.63 [-0.70, -0.57]	-0.135 [-0.143, -0.127]	+23.6 [22.2, 25.0]	-0.045 [-0.053, -0.036]
Text	10	Real-ESRGAN	-1.27 [-1.33, -1.21]	-0.194 [-0.224, -0.165]	+12.7 [11.7, 13.8]	+0.021 [0.002, 0.040]
Text	10	ResShift	-0.68 [-0.78, -0.58]	-0.214 [-0.252, -0.176]	+10.3 [9.2, 11.5]	+0.051 [0.035, 0.067]

JPEG below 30. Per image, the loss from JPEG 30 to 10 minus the loss from 50 to 30; positive means faster degradation below 30:

Method	Content	PSNR: drop 30 to 10 minus drop 50 to 30 (dB)	SSIM	LPIPS	Verdict
Bicubic	objects	+0.91 [0.83, 0.99]	+0.030 [0.028, 0.032]	+0.038 [0.032, 0.044]	supported
Bicubic	faces	+1.37 [1.31, 1.43]	+0.028 [0.027, 0.029]	+0.059 [0.056, 0.063]	supported
Bicubic	text	+0.50 [0.46, 0.55]	+0.021 [0.019, 0.023]	+0.074 [0.067, 0.081]	supported
Real-ESRNet	objects	+0.91 [0.81, 1.02]	+0.032 [0.029, 0.036]	+0.059 [0.053, 0.065]	supported
Real-ESRNet	faces	+1.38 [1.29, 1.47]	+0.022 [0.021, 0.023]	+0.050 [0.047, 0.053]	supported
Real-ESRNet	text	+1.26 [1.11, 1.40]	+0.072 [0.068, 0.076]	+0.132 [0.119, 0.146]	supported
Real-ESRGAN	objects	+0.73 [0.58, 0.89]	+0.025 [0.019, 0.031]	+0.070 [0.062, 0.080]	supported
Real-ESRGAN	faces	+1.17 [1.05, 1.29]	+0.021 [0.019, 0.022]	+0.064 [0.059, 0.069]	supported
Real-ESRGAN	text	+1.27 [1.09, 1.45]	+0.068 [0.064, 0.072]	+0.064 [0.050, 0.080]	supported
ResShift	objects	+0.69 [0.58, 0.79]	+0.028 [0.022, 0.034]	+0.079 [0.070, 0.090]	supported
ResShift	faces	+1.03 [0.95, 1.10]	+0.019 [0.017, 0.021]	+0.059 [0.054, 0.065]	supported
ResShift	text	+1.11 [0.96, 1.28]	+0.072 [0.068, 0.076]	+0.042 [0.032, 0.051]	supported

Verdicts (`results/analysis/hypothesis_checks_test.json`):

- *Generative models look sharper but lose PSNR:* supported in 11 of 12 cells. Against Real-ESRNet, LPIPS is lower, MUSIQ higher and PSNR lower, all intervals excluding zero. Exception: ResShift on text without JPEG, PSNR +0.47 [0.36, 0.58] dB.
- *They invent letters on text:* supported. CER rises while LPIPS falls: Real-ESRGAN +0.120 [0.094, 0.146], ResShift +0.023 [0.013, 0.033]. For Real-ESRGAN the rise is +0.290 [0.254, 0.327] at 24 px and -0.003 [-0.008, 0.002] at 48 px. In 111 of 384 text inputs, LPIPS ranks a generative method best while its CER is the worst of the 4.
- *They alter identity on faces:* partly supported. ArcFace similarity is lower than for Real-ESRNet: Real-ESRGAN -0.023 [-0.025, -0.021], ResShift -0.004 [-0.006, -0.003] without JPEG; -0.048 [-0.054, -0.041] and -0.045 [-0.053, -0.036] at JPEG 10. Every learned output still matches its HR face at the 0.182 threshold.
- *All models degrade sharply below JPEG 30:* supported in 12 of 12 cells for PSNR, SSIM and LPIPS. The extra PSNR loss is +0.50 to +1.38 dB. At JPEG 10, Real-ESRGAN and ResShift fall below bicubic in mean PSNR on all 3 content types; Real-ESRNet stays above it.

Dev and test agree. The best method is the same on dev and test in 55 of 60 content, JPEG and metric cells (`dev_test_consistency.json`).

4. Runtime and memory

Tesla T4, 3 warm-up runs; time per image excludes loading and saving. Peak memory: `torch.cuda.max_memory_allocated` per method, weights included.

Method	Device	Precision	Mean s/image (test)	Median s/image (test)	Peak GPU memory (MiB)	Wall time dev (s)	Wall time test (s)
Bicubic	CPU (Kaggle)	uint8 (Pillow, CPU)	0.003	0.003	n/a	17	76
Real-ESRNet	Tesla T4	float32	0.156	0.159	208	58	251
Real-ESRGAN	Tesla T4	float32	0.161	0.161	208	67	291
ResShift	Tesla T4	float32 weights, float16 autocast	0.890	0.889	2647	265	1098

Total wall time per notebook: dev 409 s, test 1718 s, seed runs 4327 s. The 4 extra ResShift seeds take 0.873 to 0.883 s per image.

Seed variability. At JPEG none and 10, the ResShift mean over images moves across 5 seeds by at most 0.047 dB PSNR, 0.0035 LPIPS, 1.45 MUSIQ, 0.0029 CER and 0.0058 ArcFace similarity. The per-image PSNR standard deviation averages 0.11 to 0.22 dB. Seed 0 ran on CUDA 12.8, seeds 1 to 4 on CUDA 13.0; the CUDA effect on dev is 0.0035 dB per image (`seed_variability_test.json`).

5. Failure analysis

Cases follow rules fixed before the test run; crops are centred on the largest error of the failing method (Section 7).

(a) LPIPS prefers wrong letters. Rule: LPIPS ranks a generative method best and its CER is the worst; largest CER gap. `text_test_075` (24 px words), JPEG 50, Real-ESRGAN: LPIPS 0.157 (best), CER 0.508 against 0.092 for bicubic. Observation: strokes are sharp, but many glyphs differ from the HR letters; Tesseract reads "circle enough winter" as "circlo Mop wintor". Likely cause: the GAN loss rewards high-contrast strokes, and LPIPS features respond to edges more than to letter identity. Evidence: the CER penalty of Real-ESRGAN peaks at 24 px (+0.290) and vanishes at 48 px. Confirming experiment: a font-size sweep in 2 px steps with the CER gap and the LPIPS-CER rank correlation per size.

(b) Altered hair texture. Rule: faces without JPEG, lowest ArcFace similarity of a generative method. `face_00591`, Real-ESRGAN: 0.912 against 0.944 for Real-ESRNet and 0.962 for bicubic; PSNR 27.84 against 30.33 dB. Observation: the crop shows dense, high-contrast hair strands in the Real-ESRGAN output. Likely cause: GAN training favours sharp texture and can change the fine structure of the hair. Evidence: the hair texture differs from HR, and Real-ESRGAN has the lowest learned-method similarity on this image at JPEG none and 50, while remaining above the match threshold. The hair crop does not establish a change of facial identity or explain the embedding difference. Confirming experiment: compare face-centred crops with fixed HR alignment, and repeat with hair masked.

(c) Generative texture below bicubic. Rule: JPEG 10, largest PSNR deficit of a generative method below bicubic. `object_0880`, ResShift: 21.31 against 24.57 dB (-3.27 dB), SSIM 0.345 against 0.561, while MUSIQ is 63.9 (HR 67.6). Observation: the HR shows fine feathers; ResShift paints a rock-like texture and a dark edge. Likely cause: the diffusion prior fills missing detail with texture that does not align with the true one. Evidence: the

deficit is 2.60 dB without JPEG and 4.56 dB at JPEG 50, so compression is probably not the main cause here. Confirming experiment: relate each object's deficit to the high-frequency energy of its HR image.

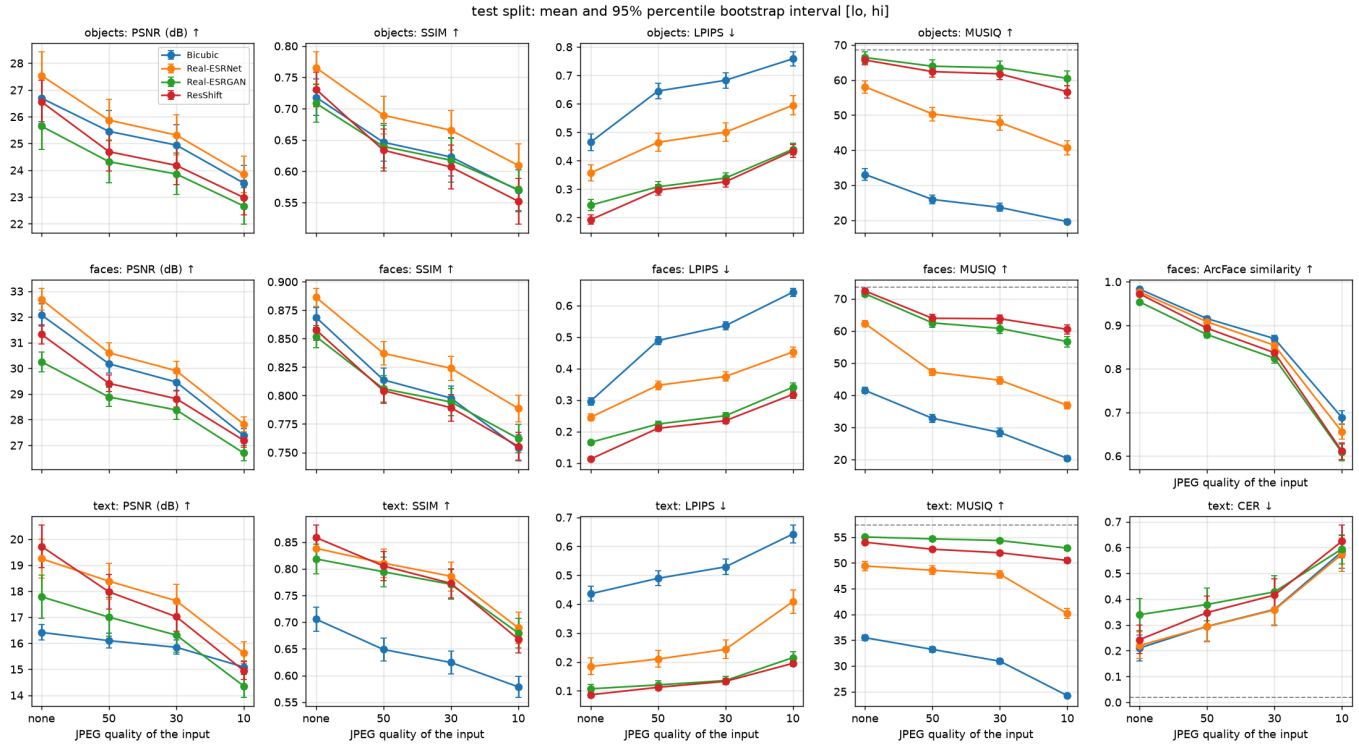
(d) Smallest font with ResShift. Rule: ResShift, 16 px, no JPEG, highest CER. `text_test_069` (random codes): CER 0.751; bicubic 0.764, Real-ESRNet 0.760, Real-ESRGAN 0.811, HR 0.040. ResShift has the best LPIPS here (0.178). Observation: letters are 4 px tall in the input; ResShift draws sharp strokes that form other characters. Likely cause: the input no longer identifies the characters, and codes give no word context. Evidence: at 16 px the mean CER without JPEG is 0.605 to 0.787 for all methods against 0.018 for HR. Confirming experiment: CER against input letter height, words and codes reported separately.

6. Limitations

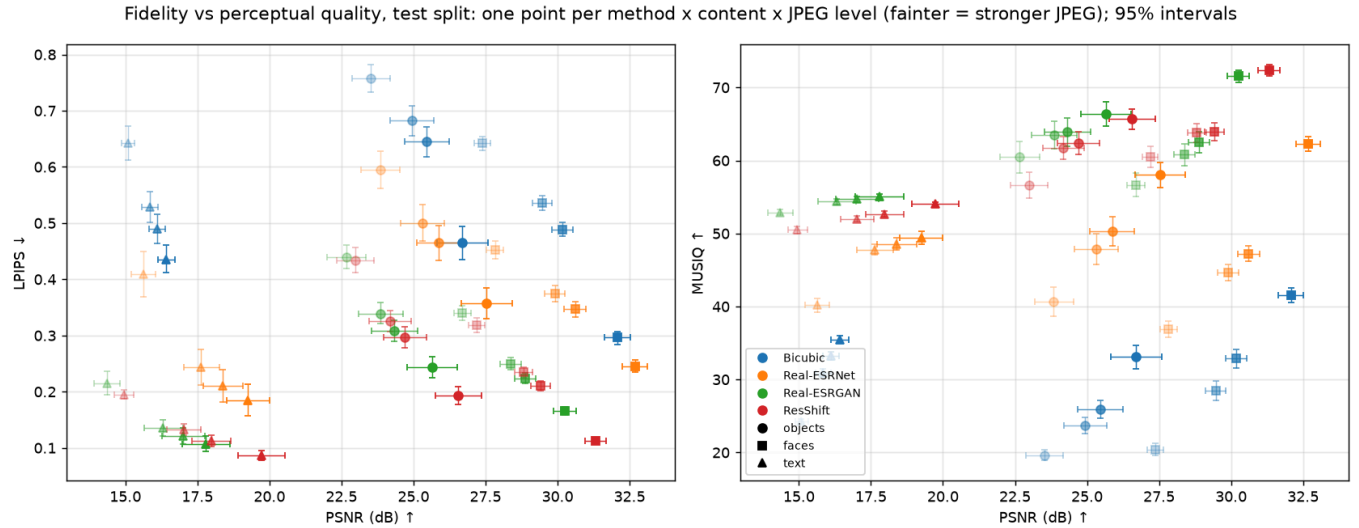
- ResShift differs from RRDBNet in architecture, training data, autoencoder, sampling and precision; gains cannot be attributed to diffusion alone.
- The ResShift v3 configuration lists FFHQ in its training data; overlap with our faces cannot be excluded.
- Intervals are pointwise, without multiple-comparison correction, and the report reads many of them.
- ResShift outputs differ between our 2 GPU environments by up to 0.032 dB per image; the main test run and the seed runs used different CUDA builds.
- The ArcFace threshold rests on 190 impostor pairs. All learned outputs match at it, so only the similarity separates the methods.
- CER uses one OCR engine and configuration. It measures readability for Tesseract, not for people.
- MUSIQ was trained on natural photographs; its scores on rendered text are hard to interpret.
- Failure crops follow the largest pixel error; in case (b) this is the hair, not the face that ArcFace uses.

7. Figures

Metrics against input JPEG quality (test, mean and 95% interval [lo, hi]; dashed: HR reference):

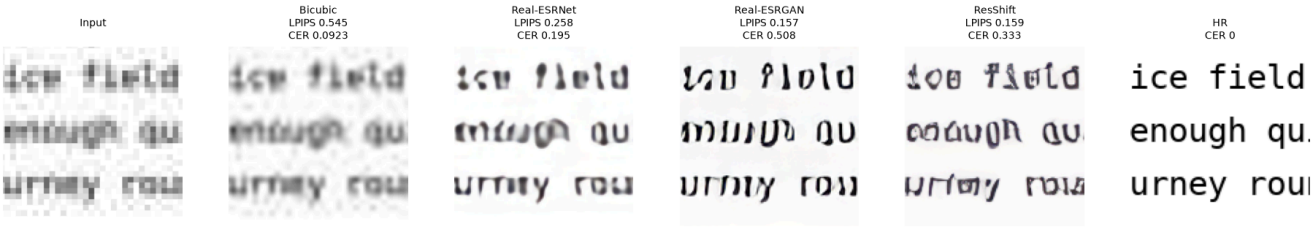


Fidelity against perceptual quality (test): one point per method, content and JPEG level, with 95% intervals.



Failure cases (a) to (d): input, the 4 methods and HR, with the values of the shown image.

Case a: text_test_075, JPEG 50, crop (235, 359, 363, 487)



Case b: face_00591, JPEG none, crop (189, 7, 317, 135)



Case c: object_0880, JPEG 10, crop (0, 148, 128, 276)



Case d: text_test_069, JPEG none, crop (0, 128, 128, 256)

